

UET University Peshawar

Abbottabad Campus



Lab Report No: 02

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Course: SE-305L Computer Communications and Networks

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Date: 18 June 2026

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Task 1: Tracing Network Route

1. Use `tracert` (Windows) to trace the route packets take to a destination:

```
C:\Users\PC_01>tracert google.com

Tracing route to google.com [172.217.17.78]
over a maximum of 30 hops:

  1    13 ms    10 ms    48 ms    192.168.10.2
  2     <1 ms   <1 ms   <1 ms    192.168.1.7
  3     1 ms    <1 ms   <1 ms    gw1.cuiatd.edu.pk [111.68.98.161]
  4     3 ms     2 ms     1 ms    172.31.68.5
  5     3 ms    43 ms    11 ms    172.31.240.134
  6     7 ms     5 ms     6 ms    172.31.240.110
  7    18 ms    18 ms    18 ms    172.31.240.101
  8    33 ms    25 ms    25 ms    172.31.240.103
  9    57 ms    24 ms    24 ms    119.63.129.237
 10    24 ms    24 ms    24 ms    110.93.252.190
 11    25 ms    24 ms    24 ms    110.93.255.138
 12    40 ms    40 ms    40 ms    72.14.204.14
 13    40 ms    42 ms    40 ms    192.178.105.155
 14    40 ms    40 ms    40 ms    192.178.105.150
 15    41 ms    41 ms    41 ms    142.251.225.200
 16    45 ms    44 ms    47 ms    192.178.98.249
 17    44 ms    44 ms    44 ms    108.170.236.209
 18    41 ms    41 ms    40 ms    ams16s30-in-f14.1e100.net [172.217.17.78]

Trace complete.
```

Questions: -

- 1) Why do packets take multiple hops?
- 2) What happens if a router on the path is down?

Answers:

- 1) Packets take multiple hops because data sent over the internet does not travel in a straight line from the source to the destination. Instead, it moves through a series of routers and networks, each forwarding the packet closer to its final destination. This happens due to:
 - **Network Infrastructure** – The internet is composed of interconnected networks, and each router directs traffic based on the best available path.
 - **Routing Protocols** – Protocols like BGP (Border Gateway Protocol) determine optimal routes for efficiency and reliability.
 - **Geographical Distance** – Packets may need to travel through multiple networks, ISPs, and backbone providers to reach the target server.
 - **Load Balancing** – Traffic is sometimes rerouted to prevent congestion and ensure smooth communication.
- 2) If a router in the path goes down, several things can happen:

- **Automatic Rerouting** – Modern networks use dynamic routing protocols like OSPF or BGP, which detect failures and find an alternate path to the destination.
- **Increased Latency** – If an alternative route is longer or congested, packets may take more time to reach their destination.
- **Packet Loss** – If no alternative route is available, packets may be dropped, resulting in timeouts or failed requests.
- **Service Disruption** – If critical routers (like those connecting different ISPs) fail and no alternate routes exist, the destination might become unreachable.

Task 2: Viewing Active Connections

1. Use `netstat` to examine active network connections: -
Windows: `netstat -an`
2. Identify open TCP and UDP connections.

```
C:\Users\PC_01>netstat -an

Active Connections

Proto Local Address           Foreign Address         State
TCP   0.0.0.0:135              0.0.0.0:0               LISTENING
TCP   0.0.0.0:445              0.0.0.0:0               LISTENING
TCP   0.0.0.0:2008             0.0.0.0:0               LISTENING
TCP   0.0.0.0:5040             0.0.0.0:0               LISTENING
TCP   0.0.0.0:5357             0.0.0.0:0               LISTENING
TCP   0.0.0.0:6045             0.0.0.0:0               LISTENING
TCP   0.0.0.0:7680             0.0.0.0:0               LISTENING
TCP   0.0.0.0:49664            0.0.0.0:0               LISTENING
TCP   0.0.0.0:49665            0.0.0.0:0               LISTENING
TCP   0.0.0.0:49666            0.0.0.0:0               LISTENING
TCP   0.0.0.0:49667            0.0.0.0:0               LISTENING
TCP   0.0.0.0:49668            0.0.0.0:0               LISTENING
TCP   0.0.0.0:49669            0.0.0.0:0               LISTENING
TCP   192.168.10.94:139        0.0.0.0:0               LISTENING
TCP   192.168.10.94:55121      40.115.3.253:443        ESTABLISHED
TCP   192.168.10.94:55137      23.10.239.251:80        CLOSE_WAIT
TCP   [::]:135                 [::]:0                  LISTENING
TCP   [::]:445                 [::]:0                  LISTENING
TCP   [::]:5357                [::]:0                  LISTENING
TCP   [::]:7680                [::]:0                  LISTENING
TCP   [::]:49664               [::]:0                  LISTENING
TCP   [::]:49665               [::]:0                  LISTENING
TCP   [::]:49666               [::]:0                  LISTENING
TCP   [::]:49667               [::]:0                  LISTENING
TCP   [::]:49668               [::]:0                  LISTENING
TCP   [::]:49669               [::]:0                  LISTENING
UDP   0.0.0.0:3702             *:0                      LISTENING
UDP   0.0.0.0:3702             *:0                      LISTENING
UDP   0.0.0.0:5050             *:0                      LISTENING
UDP   0.0.0.0:5353             *:0                      LISTENING
UDP   0.0.0.0:5355             *:0                      LISTENING
UDP   0.0.0.0:51113            *:0                      LISTENING
UDP   0.0.0.0:53331            *:0                      LISTENING
UDP   0.0.0.0:58762            *:0                      LISTENING
UDP   127.0.0.1:1900           *:0                      LISTENING
UDP   127.0.0.1:52127          *:0                      LISTENING
UDP   127.0.0.1:61153          *:0                      LISTENING
UDP   192.168.10.94:137        *:0                      LISTENING
UDP   192.168.10.94:138        *:0                      LISTENING
UDP   192.168.10.94:1900       *:0                      LISTENING
UDP   192.168.10.94:6112       *:0                      LISTENING
UDP   192.168.10.94:61152     *:0                      LISTENING
UDP   [::]:3702                *:0                      LISTENING
UDP   [::]:3702                *:0                      LISTENING
UDP   [::]:5353                *:0                      LISTENING
UDP   [::]:5355                *:0                      LISTENING
UDP   [::]:58763               *:0                      LISTENING
UDP   [::]:1900                *:0                      LISTENING
UDP   [::]:61151               *:0                      LISTENING
UDP   [fe80::918d:733b:e176:d940%12]:1900 *:0
UDP   [fe80::918d:733b:e176:d940%12]:61150 *:0

C:\Users\PC_01>
```

Questions: -

- 1) What are the common port numbers you see?
- 2) What is the difference between TCP and UDP?

Answers:

1) From the netstat -an output, some commonly seen port numbers include:

1. TCP Ports:

- 135 (RPC - Remote Procedure Call)
- 445 (SMB - Server Message Block, used for file sharing)
- 2048, 5040, 5357, 7680 (Various applications and services)
- 49664, 49665, 49666, 49667, 49668 (Dynamic/ephemeral ports assigned by Windows)
- 80 (HTTP - Web traffic)
- 443 (HTTPS - Secure web traffic)

2. UDP Ports:

- 3702 (Web Services Discovery)
- 5353 (Multicast DNS - mDNS, used for local network service discovery)
- 1900 (UPnP - Universal Plug and Play)
- 138 (NetBIOS Datagram Service, used for LAN file sharing)
- 61152, 58763 (Dynamic/ephemeral ports for temporary connections)

2) Difference Between TCP and UDP

• TCP (Transmission Control Protocol)

- Connection-oriented: Establishes a connection before data transfer.
- Reliable: Ensures all packets arrive in order and without loss.
- Uses acknowledgments (ACK) and retransmission.
- Suitable for applications like web browsing (HTTP/HTTPS), email (SMTP, IMAP), and file transfers (FTP).

- **UDP (User Datagram Protocol)**

- Connectionless: No setup required before data transfer.
- Unreliable: No guarantee of delivery or order of packets.
- Faster and lower overhead than TCP.
- Used for real-time applications like video streaming, online gaming, VoIP, and DNS lookups.

3. Add the physical cabling between devices on the workspace.

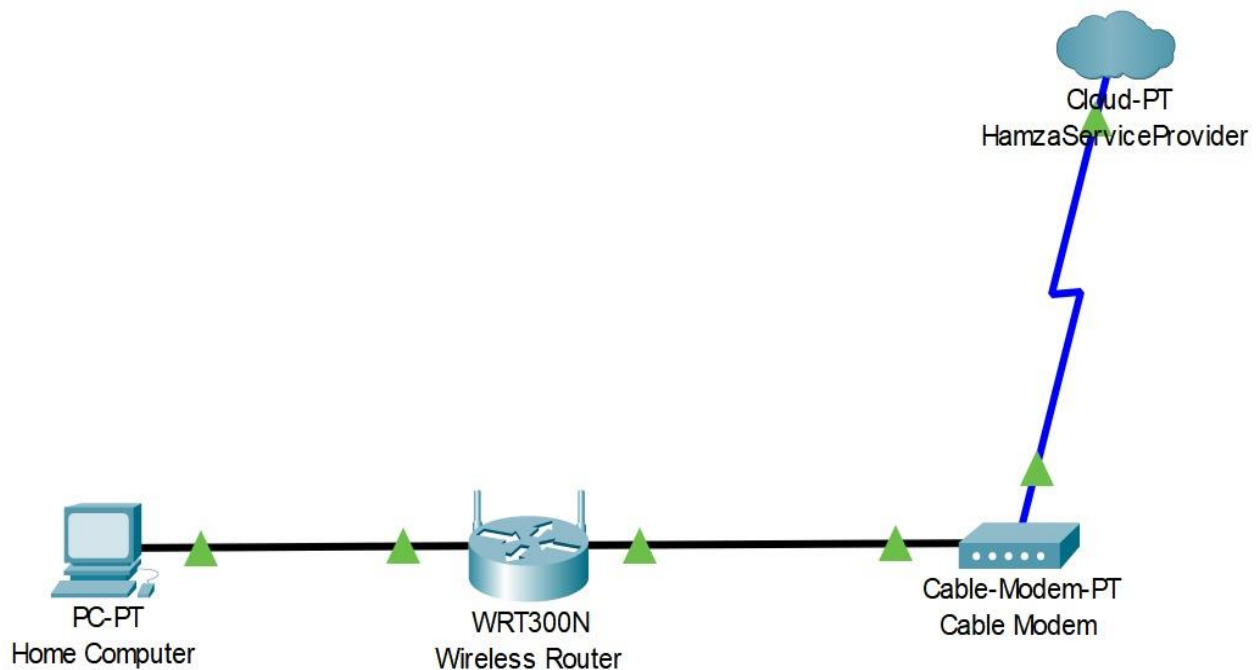


Figure 1 Basic Network